NV-CP 65-200 Centrifugal Process Pump

The NV-CP 65-200 is an end-suction, back pull-out pump for clean and slightly contaminated liquids. The rotating unit can be removed for service without disturbing the pipework or the motor.

Applications

- Cooling and heating water circuits
- Process water in food and beverage plants
- Boiler feed at moderate temperatures
- Irrigation and water transfer
- Washing and cleaning systems
- Condensate return

Design features

· Back pull-out design: the motor, bearing bracket and impeller come out as one unit.

· Replaceable wear rings on the casing and the impeller restore efficiency after years of service.

· Bearings are greased for life and sized for at least 25,000 operating hours.

· Casing drain and vent connections are tapped G 1/4 as standard.

Technical data

Hydraulic

Maximum flow	120 m³/h
Maximum head	58 m
Rated speed	2,950 rpm
Maximum working pressure	16 bar
Liquid temperature	-20 to +140 °C
Maximum viscosity	150 mm²/s

Construction

Pump casing	Ductile iron EN-GJS-400-15
Impeller	Stainless steel 1.4408, closed
Shaft seal	Single mechanical seal, SiC/SiC/EPDM
Flanges	PN 16, DN 80 suction, DN 65 discharge
Weight, bare shaft	68 kg

Performance at 2,950 rpm

Values are measured with water at 20 °C to the tolerance class 2B. The best efficiency point is at 80 m³/h.

Flow (m³/h)	0	20	40	60	80	100	120
Head (m)	58.0	57.2	55.4	52.1	47.3	40.6	31.8
Efficiency (%)	0	41	60	70	74	72	64
Shaft power (kW)	4.1	7.6	10.1	12.2	13.9	15.4	16.2
NPSH required (m)	n/a	1.6	1.9	2.4	3.1	4.2	5.8

Do not run the pump for long periods below 15 m³/h, because the liquid in the casing heats up quickly at low flow.

Motor options

Motor frame	Rated power (kW)	Supply	Efficiency class	Duty point covered
160M	11	400 V, 3-phase, 50 Hz	IE3	Up to 55 m³/h
160L	15			Up to 95 m³/h
180M	18.5		IE4	Full curve

Remote monitoring

Pumps with the optional sensor module report their operating data over Modbus RTU. A typical gateway configuration is shown below.

```
[modbus]
address  = 17
baudrate = 19200
parity   = even

; holding registers, read only
40001 = flow_m3h      ; scale 0.1
40002 = head_m        ; scale 0.1
40003 = bearing_temp_c
40005 = run_hours
```

Register 40004 is reserved and returns zero. Values are refreshed once per second.

Ordering code

Example: `NV-CP 65-200 / 150 / D / Q1`

Position	Meaning	Options
NV-CP 65-200	Pump size	Discharge DN 65, nominal impeller 200 mm
150	Motor power	110 = 11 kW, 150 = 15 kW, 185 = 18.5 kW
D	Material version	C = cast iron impeller, D = stainless steel impeller
Q1	Shaft seal	Q1 = SiC/SiC/EPDM, Q2 = SiC/carbon/FKM

Dimensions

Main dimensions of the bare-shaft pump on its baseplate are given in millimetres. Detailed drawings are available on request.

Dimension	a	f	h1	h2	Baseplate length	Baseplate width
mm	100	385	225	280	1,120	460

Standards and approvals

- Dimensions to EN 733
- Drinking water approval for the EPDM seal version
- Declaration of conformity for machinery and low-voltage equipment